EXP 6 - INDUSTRIAL HVAC CONTROL FIRMWARE: ADC TEMPERATURE ACQUISITION AND TIMER-INTERRUPT PWM FAN CONTROL

CODE:

```
const int TEMP_PIN = A0;
const int FAN_PIN = 8;

volatile byte pwmCounter = 0;
volatile byte pwmDuty = 0;

ISR(TIMER1_COMPA_vect) {
  pwmCounter++;

  if (pwmCounter >= 100) {
    pwmCounter = 0;
  }

  digitalWrite(FAN_PIN, pwmCounter < pwmDuty);
}

void setup() {
  pinMode(TEMP_PIN, INPUT);
  pinMode(FAN_PIN, OUTPUT);

  digitalWrite(FAN_PIN, LOW);

  Serial.begin(9600);
```

```
  // Configure Timer1 for a 1 ms interrupt.
  noInterrupts();

  TCCR1A = 0;
  TCCR1B = 0;
  TCNT1 = 0;

  OCR1A = 249;
  TCCR1B |= (1 << WGM12);
  TCCR1B |= (1 << CS11) | (1 << CS10);
  TIMSK1 |= (1 << OCIE1A);

  interrupts();

  Serial.println("HVAC CONTROL SYSTEM");
}

void loop() {
  int adcValue = analogRead(TEMP_PIN);

  float voltage = adcValue * (5.0 / 1023.0);
  float temperature = voltage * 100.0;

  byte duty;

  if (temperature < 25.0) {
    duty = 0;
  } else if (temperature < 30.0) {
    duty = 30;
```

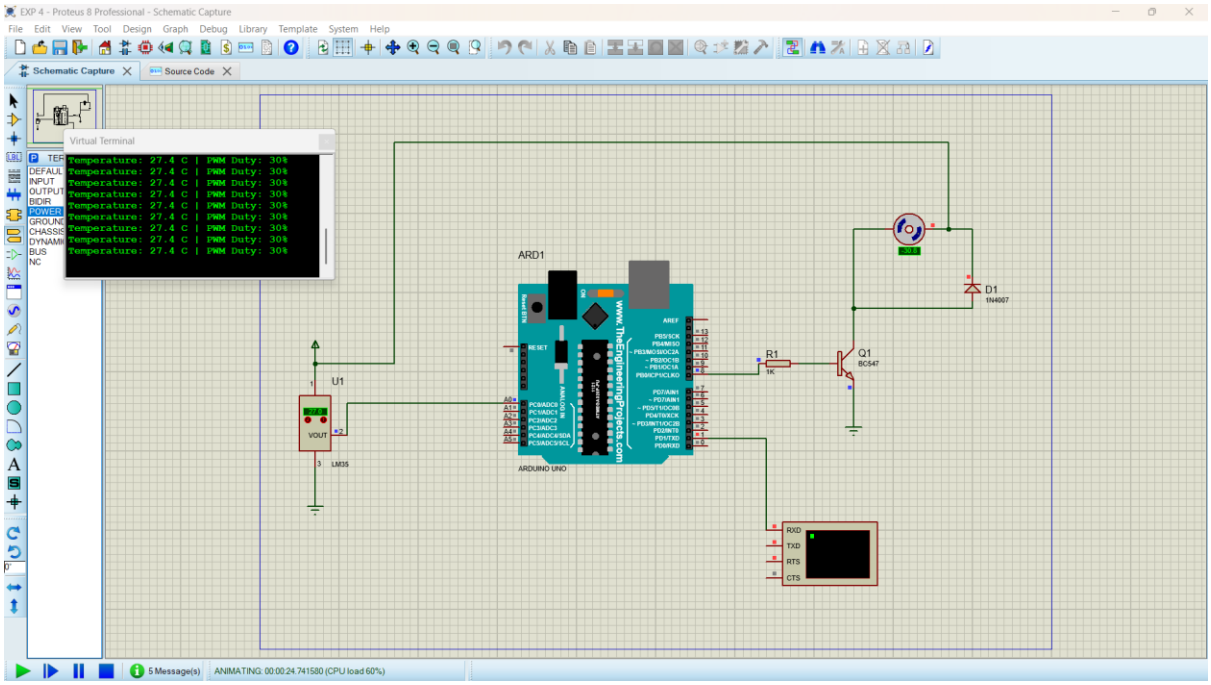
```
  } else if (temperature < 35.0) {
    duty = 60;
  } else {
    duty = 90;
  }

  noInterrupts();
  pwmDuty = duty;
  interrupts();

  Serial.print("Temperature: ");
  Serial.print(temperature, 1);
  Serial.print(" C | PWM Duty: ");
  Serial.print(duty);
  Serial.println("%");

  delay(1000);
}
```

OUTPUT:


EXP 4 - Proteus 8 Professional - Schematic Capture
Schematic Capture
Source Code
Virtual Terminal
Temperature: 27.4 C | PWM Duty: 30%
ARD1
U1
LM35
R1
1K
Q1
BC547
D1
1N4007
ARDUINO UNO
RXD
TXD
RTS
CTS
5 Message(s)
ANIMATING: 00:00:24.741580 (CPU load 60%)